Engaging diverse communities: the ATRIUM project's participatory research initiatives

Niccolucci, Ginevra
ginevra.niccolucci@prisma-cultura.it
Prisma Cultura S.r.l. - Società Benefit, Italy

Prandoni, Claudio
claudio.prandoni@ariadne-research-infrastructure.eu
ARIADNE Research Infrastructure AISBL

Niccolucci, Franco
franco.niccolucci@gmail.com
ARIADNE Research Infrastructure AISBL

Geser, Guntram
guntram.geser@gmail.com
ARIADNE Research Infrastructure AISBL

1. The ATRIUM project

Non-professional communities play a crucial role in collaborative cultural heritage research projects. The ATRIUM project seeks to engage with various groups, including metal detectorists, rock art archivists, Nuragic history enthusiasts, Linked Open Data networks, deaf and blind individuals, to enhance accessibility. This presentation highlights the ongoing efforts in participatory research, emphasising the involvement of these communities in activities ranging from methodological discussions to data collection, analysis and publication.

The ATRIUM (Advancing fronTier Research In the arts and hUManities) project aims to bridge leading research infrastructures in arts and humanities (DARIAH), archaeology (ARIADNE), languages (CLARIN), and open scholarly communication in the social sciences and humanities (OPERAS).

“Arts and Humanities” encompass a diverse range of disciplines and communities with varying epistemological and methodological foundations. For instance, an archaeologist and an art historian studying a Mycenaean fresco will employ distinct goals and approaches, just as a literary scholar and a linguist will interpret a textual corpus differently. Research infrastructures in this domain must cater to a wide array of stakeholders and offer services that transcend discipline-specific boundaries.

The ATRIUM project addresses this heterogeneity by adopting a dual approach: it aims to make significant contributions to the consolidation and expansion of data management and processing services. Concurrently, ATRIUM facilitates access to a broad spectrum of essential text, image, and sound-based services beneficial to various disciplines within the Arts and Humanities, covering all phases of the research data lifecycle (creation, processing, analysis, preservation, access, and reuse).

2. ATRIUM for non-professional researchers

Within the project, a research strand is dedicated to the analysis of the ways of engaging non-professional communities. This activity involves setting up **participatory research initiatives** through external collaborations with active communities and their organisations. The dual purpose is to foster the interest of non-professional communities in curiosity-driven activities and to gather feedback on the implementation of ATRIUM workflows, service and demonstrators.

One such community is formed by the **metal detectorists**, active in several Northern and Eastern European countries, notably the UK, Netherlands, and Denmark, where the activity is legal with landowner permission and reporting requirements. In other countries, licensing may be required. Participants are typically hobbyists focusing on non-ferrous metals in the open spaces. In England and Wales, a national recording scheme, the **Portable Antiquities Scheme (PAS)** was established in 1997 and over 1.5 million finds made by members of the public have now been recorded. The initiative is currently coordinated by the British Museum. In the Czech Republic, metal detecting has gained popularity over the past 20-30 years, but legal conditions are restrictive. Current legislation allows only authorized institutions to conduct archaeological fieldwork, including metal detecting, with amateurs participating under their supervision. In 2020, the Portal of Amateur Collaborators and Register of Individual Finds of the Archaeological Map of the Czech Republic (**AMCR-PAS**) was launched, providing an official digital scheme for recording activities and finds. Other platforms are available and used in other countries: such as

DIME in Denmark and PAN, the Portable Antiquities of the Netherlands.

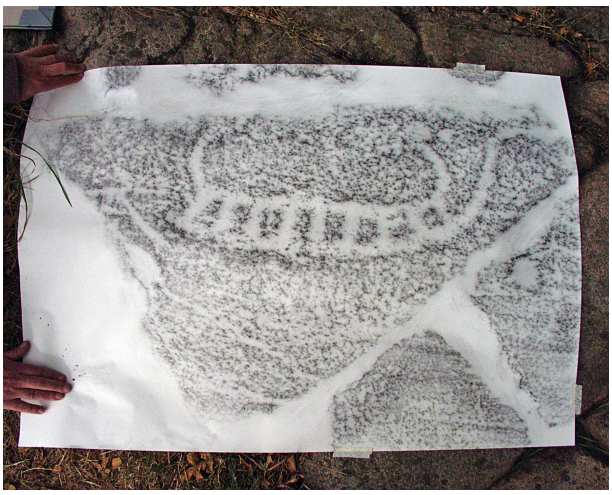

Figure 1 – A silver Roman denarius of Augustus (PAS)

The **Nuraghe network NURNET** in Sardinia aims to disseminate the Nuragic civilization, highlighting its uniqueness. The goal is to create a geoportal of sites from the prehistoric and protohistoric periods through public participation, fostering a community to discuss the archaeological landscape and Sardinian society. The content of the Nurnet geoportal is provided via crowdsourcing after being reviewed by professionals.

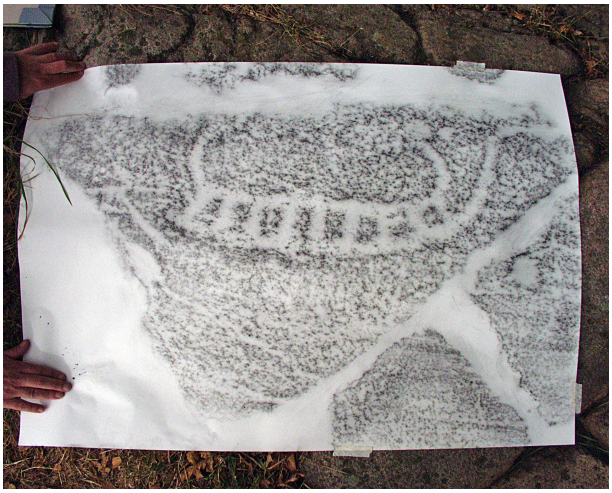

Figure 2 – Nuraghe Nieddu from Codrongianos (Nurnet)

The **Swedish Rock Art Research Archives** (SHFA) at the University of Gothenburg is a research outcome of many surveyors, primarily private citizens, who discover and document rock carvings. The archive largely includes data from professionals, with some private collections from semi-professionals, most of the interaction with and work by non-professionals focuses now on students and courses.

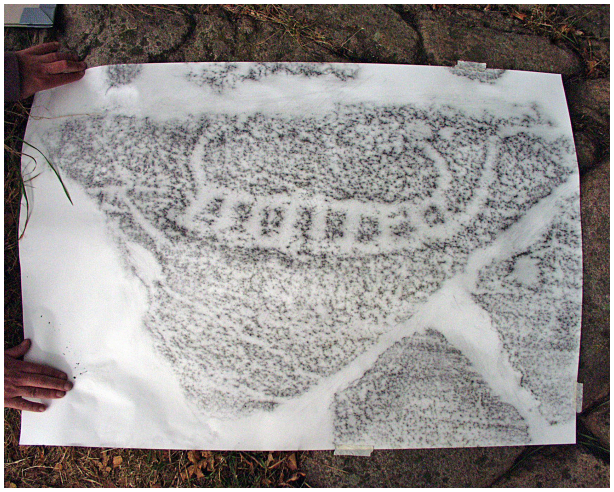

Figure 3 – Graffiti of a ship (SHFA)

The **Pelagios Network** is a community of researchers, scientists, and curators using Linked Open Data (LOD) to study the past. As a free and open association run by and for its members, the network focuses on activities such as annotation, gazetteers, registering and visualization of geospatial information. It has developed tools like *Recogito* for annotating place information, *Peripleo* for linking Linked Data resources, the *World Historical Gazetteer* for gathering placename data, and *PeriodO* for scholarly definitions of historical periods. These resources are also available to non-professional communities engaging in cultural heritage mapping projects.

3. ATRIUM and accessibility

To improve access to art and heritage for **citizens with special needs**, the ATRIUM project involves the **deaf and blind communities in Italy**. The deaf community using Italian Sign Language (LIS) comprises approximately 40,000 people, with an estimated 100,000 signers including hearing individuals. Across the EU, there are an estimated 750,000 deaf individuals using various sign languages.

The Italian Union of the Blind and Partially Sighted (UICI) advocates for equal rights and accessibility. Italy has over 122,000 blind and 1.5 million partially sighted individuals. The European Blind Union (EBU) represents 42 countries, highlighting the broader European context with an estimated 30 million people. Deaf and blind people are often

excluded from enjoying cultural heritage due to a lack of dedicated content.

Within the ATRIUM project, ARIADNE RI will create a glossary in LIS (Italian Sign Language) and IS (International Sign) for archaeology. The aim of **AGLAIA** (**ARIADNE Glossary in LIS And IS for Archaeology**) is to propose a shared foundation in LIS and IS for the most commonly used terms in guided tours of museums and archaeological sites. The resulting glossary will be made freely available online and as a downloadable PDF. ARIADNE will also provide technological support to its members who wish to develop a similar glossary in their own national sign languages (BSL, LSF, DGS, LSE, etc.).

4. The ATRIUM impact

To gather information on the state of the art, the needs, and the requirements of the non-professional communities involved in ATRIUM, the project created a dedicated **survey**. The information collected facilitates the creation of standardised workflows that enhance data and service interoperability across Research Infrastructures, supported by demonstrators addressing real-world, research-driven questions in the arts and humanities.

The contributions of non-professional communities will significantly benefit the ATRIUM project. All communities share a strong interest in cultural heritage, whether through hobbies, professional collaboration, or advocacy for accessibility. Their diverse backgrounds offer added value and opportunities for comparison. Each community collects relevant data while adhering to specific legal requirements, ensuring ethical and responsible practices. The survey highlights common tools of interest, including metadata extraction, automatic image recognition, and interactive visualization tools, indicating a need for advanced technological support.

All communities express a willingness to collaborate with the ATRIUM project, underscoring the potential for mutual benefit and enhanced services. By leveraging their contributions, the ATRIUM project can improve services, accessibility, and foster a more inclusive and participatory research environment. The common characteristics identified across these communities suggest a strong potential for collaborative initiatives and the development of tailored services to meet their specific needs.

5. References to the home pages of the above-mentioned initiatives

ATRIUM https://atrium-research.eu/
DARIAH https://www.dariah.eu/
ARIADNE https://www.ariadne-research-infrastructure.eu/
CLARIN https://www.clarin.eu/
OPERAS https://operas-eu.org/
Portable Antiquities Scheme https://finds.org.uk/
AMCR PAS https://amcr-info.aiscr.cz/?page=pas (in Czech)
DIME https://www.metaldetektorfund.dk/information/?om (in Danish)
PAN https://portable-antiquities.nl/pan/#/public (in Dutch)
NURNET https://www.nurnet.net/ (in Italian)
NURNET geoportal https://nurnet.crs4.it/nurnet-geo/
PELAGIOS https://pelagios.org/
SHFA https://www.gu.se/en/shfa

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Figure 2: By Gianni Careddu, own work, CC BY-SA 4.0
Figure 3: By Gerhard Milstreu, SHFA (2005), CC BY 4.0